

Convex Cones and Generalized Interiors

Bauschke & Combettes, 2017 (Chapter 6)

Lecture on Convex Analysis

Convex Analysis and Monotone Operator Theory

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What is a Convex Cone?

Definition (Convex Cone)

A set $K \subseteq \mathcal{H}$ is a **cone** if

$$(\forall \lambda \in \mathbb{R}_+) \quad \lambda K \subseteq K$$

where $\mathbb{R}_+ = [0, \infty)$.

Interpretation: If $x \in K$, then all positive scalar multiples λx are also in K .

[Properties of Cones]

- A cone always contains the origin: $0 \in K$
- A cone closed under nonnegative combinations: if $x_i \in K$ and $\lambda_i \geq 0$, then $\sum_i \lambda_i x_i \in K$
- The intersection of cones is a cone
- The closure of a cone is a cone

Example: The Positive Orthant

Example (Positive Orthant)

The positive orthant in $\mathbb{R}^n$ is

$$\mathbb{R}_+^n = \{x = (x_1, \dots, x_n) : x_i \geq 0 \text{ for all } i\}$$

This is a closed, pointed, solid convex cone.

Example (Half-Space Cone)

Given $u \in \mathcal{H} \setminus \{0\}$, the set

$$K = \{x \in \mathcal{H} : \langle x | u \rangle \leq 0\}$$

is a solid convex cone, and it is not pointed if and only if $\dim \mathcal{H} > 1$.

$K = \mathbb{R}_+^2$ is self-dual: $K^\ominus = -K$, $K^\oplus = K$

The cone K



The polar cone $K^\ominus$



Cone Generated by a Set

Definition

Given a set $C \subseteq \mathcal{H}$, the **cone** (or **conic hull**) generated by C is

$$\text{cone}(C) = \left\{ \sum_{i \in I} \lambda_i x_i : I \text{ finite}, \lambda_i \in \mathbb{R}_+, x_i \in C \right\}$$

[6.2] Let $D = \mathbb{R} \cdot C$. Then:

- ① $\text{cone}(C) \subseteq \text{cone}(D) = D$
- ② If C is a cone, then $\text{cone}(C) = C$
- ③ $\overline{\text{cone}(C)} = \text{cone}(\overline{C})$

Cone Generated by v_1, v_2 : $K = \text{cone}\{v_1, v_2\}$



Relationships Between Cones and Convex Sets

[6.3] Let C be a subset of $\mathcal{H}$. Then:

- (i) If C is a cone, then C is convex if and only if $C + C \subseteq C$.
- (ii) If C is convex and $0 \in C$, then C is a cone if and only if $C + C \subseteq C$.

[6.4] Let C be a nonempty convex subset of $\mathcal{H}$. Then:

- (i) $\text{span}(C) = \text{cone}(C) - \text{cone}(C) = \text{cone}(C) + \text{cone}(-C)$
- (ii) If $C = -C$, then $\text{span}(C) = \text{cone}(C)$

Key insight: The cone generated by C gives us the smallest cone containing C .

Pointed and Solid Cones

Definition (Pointed and Solid Cones)

Let K be a convex cone in $\mathcal{H}$. Then:

- K is **pointed** if $K \cap (-K) \subseteq \{0\}$
- K is **solid** if $\text{int}(K) \neq \emptyset$
- The only pointed linear subspace is $\{0\}$
- $\mathcal{H}$ is the only solid linear subspace
- The positive orthant $\mathbb{R}_+^n$ is both pointed and solid (for $n \geq 1$)

Example (Half-Space)

The half-space $K = \{x \in \mathcal{H} : \langle x | u \rangle \leq 0\}$ is:

- Solid (for any $u \neq 0$)
- Pointed if and only if $\dim \mathcal{H} = 1$

Pointed vs Non-Pointed Cones

Standard Interior

Recall that the interior of a set C is:

$$\text{int}(C) = \{x \in C : (\exists \rho \in \mathbb{R}_{++}) B(\rho; x) \subseteq C - x\}$$

For convex sets, we have the hierarchy:

$$\text{int}(C) \subseteq \text{core}(C) \subseteq \text{sri}(C) \subseteq \text{ri}(C) \subseteq \text{qri}(C) \subseteq C$$

Definition (Generalized Interiors)

Let C be a convex subset of $\mathcal{H}$:

- **Core:** $\text{core}(C) = \{x \in C : \text{cone}(C - x) = \mathcal{H}\}$
- **Strong relative interior:** $\text{sri}(C) = \{x \in C : \text{cone}(C - x) = \overline{\text{span}(C - x)}\}$
- **Relative interior:** $\text{ri}(C) = \{x \in C : \text{cone}(C - x) = \text{span}(C - x)\}$
- **Quasi-relative interior:** $\text{qri}(C) = \{x \in C : \overline{\text{cone}(C - x)} = \overline{\text{span}(C - x)}\}$

Key Properties of Generalized Interiors

[6.10] Let C be a nonempty convex subset of $\mathcal{H}$ such that $C = -C$. Then:

- (i) $0 \in \text{ri}(C)$
- (ii) If $\text{span}(C)$ is closed, then $0 \in \text{sri}(C)$

For any convex set C , we always have:

$$\text{int}(C) \subseteq \text{core}(C) \subseteq \text{sri}(C) \subseteq \text{ri}(C) \subseteq \text{qri}(C) \subseteq C$$

Each inclusion can be strict (see Example 6.11).

Examples of Generalized Interiors I

Example (6.11)

- (i) A proper closed linear subspace V has $\text{core}(V) = \emptyset$ but $\text{sri}(V) = V$.
- (ii) In infinite-dimensional spaces, consider the set

$$C = \left\{ \sum_{n \in \mathbb{N}} \xi_n e_n : (\forall n) |\xi_n| \leq \frac{1}{4^n} \right\}$$

Then $\text{cone}(C) \neq \overline{\text{span}(C)} = \mathcal{H}$, so $\text{core}(C) = \emptyset$.

- (iii) However, $0 \in \text{sri}(C)$ (with stronger growth condition) when the span is closed.

Examples of Generalized Interiors II

The key difference between $\text{ri}(C)$ and $\text{core}(C)$ is whether we require the closure:

$$x \in \text{core}(C) \Rightarrow \text{cone}(C - x) = \mathcal{H}$$

$$x \in \text{sri}(C) \Rightarrow \text{cone}(C - x) = \overline{\text{span}(C - x)}$$

In infinite dimensions, this distinction matters significantly!

Polar and Dual Cones

Definition (Polar Cone)

Given a set K in $\mathcal{H}$, the **polar cone** is

$$K^{\ominus} = \{u \in \mathcal{H} : (\forall x \in K) \langle u | x \rangle \leq 0\}$$

Definition (Dual Cone)

The **dual cone** is

$$K^{\oplus} = \{u \in \mathcal{H} : (\forall x \in K) \langle u | x \rangle \geq 0\} = -K^{\ominus}$$

Relationship: $K^{\oplus} = -K^{\ominus}$ (the dual cone is the negative polar).

- If K is a cone, then $K^{\ominus}$ and $K^{\oplus}$ are both closed convex cones
- $(K^{\ominus})^{\ominus} = \overline{\text{cone}(K)}$ (bipolar theorem)
- $\mathbb{R}_+^n$ is self-dual: $(\mathbb{R}_+^n)^{\oplus} = \mathbb{R}_+^n$

Ice-cream cone K : $K^{\ominus} = -K$, so $K^{\oplus} = -K^{\ominus} = K$ (self-dual)

Properties of Polar and Dual Cones

[6.24] Let K be a nonempty closed convex cone in $\mathcal{H}$, let $x \in \mathcal{H}$, and let $p \in \mathcal{H}$. Then

$$p = P_K x \Leftrightarrow [p \in K, x - p \in K^\oplus]$$

Equivalently,

$$p = P_K x \Leftrightarrow [p \in K, x - p \in K^\ominus]$$

Interpretation: The projection onto K is characterized using the dual cone.

Theorem (6.30 - Moreau)

Let K be a nonempty closed convex cone in $\mathcal{H}$ and let $x \in \mathcal{H}$. Then:

- ① $x = P_K x + P_{K^\oplus} x$
- ② $P_K x \perp P_{K^\oplus} x$
- ③ $\|x\|^2 = d_K^2(x) + d_{K^\oplus}^2(x)$

Self-Dual Cones

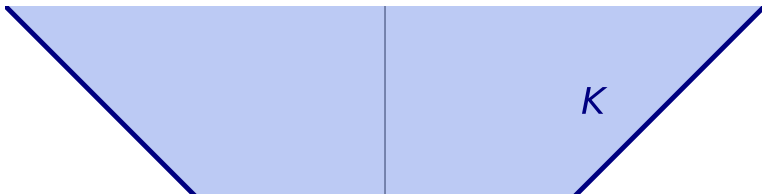
Definition

A cone K is **self-dual** if $K^\oplus = K$ (or equivalently, $K^\ominus = -K$).

Example

- The positive orthant $\mathbb{R}_+^n$ is self-dual
- The ice-cream (second-order) cone $\{(x, y) \in \mathbb{R}^{n-1} \times \mathbb{R} : y \geq \|x\|\}$ is self-dual
- The Lorentz cone is self-dual

Ice-cream cone K : $K^\ominus = -K$, so $K^\oplus = -K^\ominus = K$ (self-dual)



Projection onto Cones

[6.28] Let K be a nonempty closed convex cone in $\mathcal{H}$, let $x \in \mathcal{H}$, and let $p \in \mathcal{H}$. Then $p = P_K x$ if and only if

$$p \in K, \quad x - p \in K^\ominus, \quad \langle x - p \mid p \rangle = 0$$

Key property: The orthogonality condition $\langle x - p \mid p \rangle = 0$ is crucial.

Example (6.29)

Let $\mathcal{H} = \ell^2(I)$, $K = \ell_+^2(I)$, and $x = (\xi_i)_{i \in I}$. Then

$$P_K x = (\max\{\xi_i, 0\})_{i \in I}$$

This is the componentwise positive part operation.

Tangent Cone

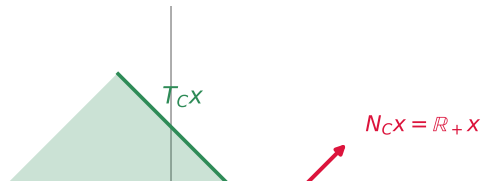
Definition (Tangent Cone)

Let C be a nonempty convex subset of $\mathcal{H}$ and let $x \in \mathcal{H}$. The **tangent cone** to C at x is

$$T_C x = \begin{cases} \overline{\text{cone}(C - x)} = \bigcup_{\lambda \in \mathbb{R}_{++}} \lambda(C - x) & \text{if } x \in C; \\ \emptyset & \text{otherwise} \end{cases}$$

- If $x \in C$, then $T_C x$ is a nonempty closed convex cone containing 0
- At an interior point: $T_C x = \mathcal{H}$
- At a boundary point, $T_C x$ captures directions we can move into C

Tangent and normal cone to the unit ball at a boundary point



Normal Cone

Definition (Normal Cone)

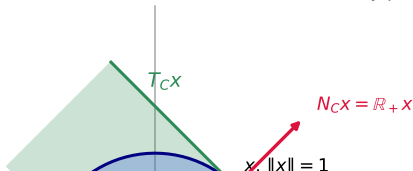
Let C be a nonempty convex subset of $\mathcal{H}$ and let $x \in \mathcal{H}$. The **normal cone** to C at x is

$$N_C x = \{u \in \mathcal{H} : (\forall y \in C) \langle u | y - x \rangle \leq 0\}$$

Interpretation: $u \in N_C x$ means u points “outward” from the tangent directions.

- If $x \notin C$, then $N_C x = \emptyset$ (definition extends to: $N_C x = \emptyset$ if $x \notin C$, or alternatively, $N_C x = \{u : \text{the hyperplane } \{y : \langle u | y - x \rangle = 0\} \text{ supports } C\}$)
- If $x \in C$, then $N_C x$ is a nonempty closed convex cone
- At an interior point: $N_C x = \{0\}$

Tangent and normal cone to the unit ball at a boundary point



Relationship Between Tangent and Normal Cones

[6.44] Let C be a nonempty convex subset of $\mathcal{H}$ and let $x \in C$. Then:

- ① $T_C^\oplus x = N_C x$ and $N_C^\oplus x = T_C x$
- ② If $x \in \text{core}(C)$, then $T_C x = \mathcal{H} \Leftrightarrow N_C x = \{0\}$

[6.45] Let C be a convex subset of $\mathcal{H}$ such that $\text{int}(C) \neq \emptyset$ and let $x \in C$. Then:

$$x \in \text{int}(C) \Leftrightarrow T_C x = \mathcal{H} \Leftrightarrow N_C x = \{0\}$$

Key insight: The tangent and normal cones are orthogonal complements!

Examples of Tangent and Normal Cones

Example (6.39 - Unit Ball)

For $C = B(0; 1) = \{x \in \mathcal{H} : \|x\| \leq 1\}$ and $x \in C$ with $\|x\| = 1$:

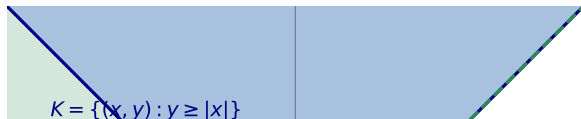
$$T_C x = \{y \in \mathcal{H} : \langle y | x \rangle \leq 0\}, \quad N_C x = \mathbb{R}_+ x$$

Example (6.40 - Ice-Cream Cone)

For the ice-cream cone $K = \{(x, y) \in \mathbb{R}^{n-1} \times \mathbb{R} : y \geq \|x\|\}$ at $x_0 = (x_0, y_0)$ with $y_0 > \|x_0\|$:

$$T_K x_0 = \{(x, y) : y \geq \|x\|\}, \quad N_K x_0 = \mathbb{R}_+ \cdot \frac{(x_0, -\|x_0\|)}{\sqrt{2\|x_0\|^2 + 1}}$$

Tangent and normal cone to K at the boundary point $x_0 = (1, 1)$



Characterization of Interior Points

Corollary (6.46)

Suppose that $\mathcal{H}$ is finite-dimensional, let C be a nonempty convex subset of $\mathcal{H}$, and let $x \in C$. Then the following are equivalent:

- ① $x \in \text{int}(C)$
- ② $T_C x = \mathcal{H}$
- ③ $N_C x = \{0\}$

[6.47] Let C be a nonempty closed convex subset of $\mathcal{H}$, and let x and p be points in $\mathcal{H}$. Then $p = P_C x$ if and only if $x - p \in N_C p$.

Key insight: The projection characterization uses the normal cone in a fundamental way.

Python Example: Projection onto Positive Orthant

```
import numpy as np

def project_onto_positive_orthant(x):
    """
    Project a vector  $x$  onto the positive orthant  $R_+^n$ .

    For convex cones:  $P_K(x) = (\max(x_i, 0))_{i=1}^n$ 
    """
    return np.maximum(x, 0)

# Example
x = np.array([2, -1, 3, -0.5, 1])
proj_x = project_onto_positive_orthant(x)
print(f"Original: {x}")
print(f"Projection onto  $R_+^n$ : {proj_x}")

# Verify:  $x - \text{proj}_x$  should be in  $K^* = \{u: \langle u, y \rangle \geq 0 \text{ for } y \text{ in } K\}$ 
residual = x - proj_x # [-2, -1, 0, -0.5, 0] (all  $\leq 0$  dual cone)
print(f"Residual: {residual}")
```

Output:

- Original: [2, -1, 3, -0.5, 1]
- Projection: [2, 0, 3, 0, 1]
- Residual: [0, -1, 0, -0.5, 0] (in dual cone)

Python Example: Polar Cone

```
import numpy as np

def check_in_polar_cone(x, u):
    """
    Check if u is in the polar cone  $K^\circ = \{u: \langle u, x \rangle \leq 0 \text{ for all } x \text{ in } K\}$ 
    For  $K = \mathbb{R}_+^n$ , this is  $\mathbb{R}_-^n$  (the negative orthant).
    """
    return np.dot(u, x) <= 1e-10 # numerical tolerance

# Example:  $K = \mathbb{R}_+^2$ 
K_x = np.array([1, 2]) # in  $K = \mathbb{R}_+^2$ 

u1 = np.array([-1, -1]) # in  $K^\circ$  (polar)
u2 = np.array([1, 1]) # NOT in  $K^\circ$ 

print(f"x = {K_x} in K =  $\mathbb{R}_+^2$ ")
print(f"u1 = {u1}:  $\langle u1, x \rangle = \{np.dot(u1, K_x)\} \leq 0$ ? {check_in_polar_cone(K_x, u1)}")
print(f"u2 = {u2}:  $\langle u2, x \rangle = \{np.dot(u2, K_x)\} \leq 0$ ? {check_in_polar_cone(K_x, u2)}")

# Verify:  $K^\circ = -K$  (self-dual property of  $\mathbb{R}_+^n$ )
print(f"\nFor K =  $\mathbb{R}_+^n$ :  $K^\circ = -K$  (self-dual)")
```

Output:

- $u1 = [-1, -1]$: $\langle u1, x \rangle = -3 \leq 0$? True
- $u2 = [1, 1]$: $\langle u2, x \rangle = 3 \leq 0$? False

Python Example: Tangent and Normal Cones

```
import numpy as np

def tangent_cone_ball(x, radius=1.0):
    """
    For  $C = B(0, r)$ , at point  $x$  with  $\|x\| \leq r$ :
     $T_C(x) = \{y : \langle y \mid x \rangle \leq 0\}$  if  $\|x\| = r$  (boundary)
     $H$  if  $\|x\| < r$  (interior)
    """
    norm_x = np.linalg.norm(x)
    if np.isclose(norm_x, radius):
        return "T_C(x) = {y : <y | x> <= 0}" # half-space
    else:
        return "T_C(x) = H (entire space)"

def normal_cone_ball(x, radius=1.0):
    """
    For  $C = B(0, r)$ , at point  $x$  with  $\|x\| \leq r$ :
     $N_C(x) = R_+ * x$  if  $\|x\| = r$  (boundary)
     $\{0\}$  if  $\|x\| < r$  (interior)
    """
    norm_x = np.linalg.norm(x)
    if np.isclose(norm_x, radius):
        return f"N_C(x) = R_+ * {x/norm_x}" # ray
    else:
        return "N_C(x) = {0}"
```

Example on unit ball boundary

Fundamental Decomposition

Theorem (Projection Decomposition)

Let K be a nonempty closed convex cone. Then for every $x \in \mathcal{H}$:

$$x = P_K x + P_{K^\oplus} x$$

with

$$P_K x \perp P_{K^\oplus} x \quad \text{and} \quad \|x\|^2 = \|P_K x\|^2 + \|P_{K^\oplus} x\|^2$$

Interpretation: Every vector can be uniquely decomposed into a component in the cone and one in the dual cone, with orthogonality.

Corollary (6.31)

Let K be a nonempty closed convex cone in $\mathcal{H}$, and let $x \in \mathcal{H}$. Suppose that $y \in K$ and $z \in K^\oplus \setminus \{P_{K^\oplus} x\}$ satisfy $x = y + z$. Then:

- ① $\|P_K x\| \leq \|y\|$ and $\|P_{K^\oplus} x\| \leq \|z\|$
- ② $\langle y \mid z \rangle = \langle P_K x \mid P_{K^\oplus} x \rangle = 0$

Summary of Key Properties

Property	Cone K	Dual/Polar
Nonempty	Yes	Yes
Closed	Yes	Yes
Convex	Yes	Yes
Contains 0	Yes	Yes
Self-Duality	$K^{\oplus} = K$ for some cones	
Bipolar	$(K^{\ominus})^{\ominus} = \overline{\text{cone}(K)}$	

- **Positive Orthant:** Pointed, solid, self-dual
- **Half-space:** Solid, not pointed (unless $\dim = 1$)
- **Linear subspace:** Both a cone and its own polar (up to sign)
- **Ice-cream cone:** Pointed, solid, self-dual

Applications and Connections I

- **Optimization:** Convex cones appear naturally in conic programming
 - Linear conic programs: minimize $\langle c \mid x \rangle$ subject to $Ax \in K$
 - KKT conditions involve normal cones
- **Duality:** Lagrange multipliers lie in the dual cone
 - Primal: minimize $f(x)$ subject to $x \in K$
 - Dual involves $K^\oplus$
- **Monotone Operator Theory:** Normal cones define subdifferentials
 - Subdifferential $\partial f(x) = N_{\text{epi}(f)}(x, f(x))$
 - Used in proximal algorithms
- **Matrix Optimization:** Semidefinite cones $\mathbb{S}_+^n$
 - Convex cones in the space of symmetric matrices
 - Self-dual: $(\mathbb{S}_+^n)^\oplus = \mathbb{S}_+^n$

Summary

- ① **Convex Cones:** Fundamental objects containing 0 and closed under positive scaling
- ② **Generalized Interiors:** Multiple notions adapted for non-Euclidean spaces (core, sri, ri, qri)
- ③ **Polar & Dual Cones:** Orthogonal complements in the cone setting; crucial for optimization
- ④ **Tangent & Normal Cones:** Characterize boundary behavior; dual pairs via polarity
- ⑤ **Key Result:** Projection decomposition $x = P_K x + P_{K^\oplus} x$ with orthogonality

Next Chapter: Support functions and polar sets generalize these ideas further.

- **Bauschke, H. H., & Combettes, P. L.** (2017). *Convex Analysis and Monotone Operator Theory in Hilbert Spaces*. Springer, 2nd edition. Chapter 6.
- **Boyd, S., & Vandenberghe, L.** (2004). *Convex Optimization*. Cambridge University Press. (Chapters on convex sets and cones)
- **Rockafellar, R. T.** (1970). *Convex Analysis*. Princeton University Press. (Classic reference on convex analysis)

Thank you!